

CAPSTONE PROJECT REPORT

Order to Cash (O2C): The Order to
Cash
(O2C) Process End to End Using
SAP SD

Prepared By: Shubhang Jha

Roll No: 23051058

Program: B.Tech CSE 2027

Institution Name: Kalinga Institute of Industrial
Technology, Bhubaneswar

Name	Shubhang Jha
Roll Number	23051058
Program	B.Tech CSE
Topic	O2C (Order to Cash)
Approach	End-to-End Scenario

Abstract:

This complete capstone project explains the process of implementing the Order-to-Cash (O2C) cycle in the SAP SD module. The project is an example of how complicated business processes can be integrated into one ERP system, starting from pre-sales process to financial clearing processes. In the case study of "TechVision Enterprises Pvt. Ltd." . This report explains the process of configuring master data, conducting transactions, and integrating processes in SAP MM, FI, and WM modules.

Main Objectives:

- To create an automated process for sales that reduces human error.
- To incorporate real-time tracking of inventory and finances throughout the supply chain.
- To adhere to Indian tax (GST) regulations in the SAP system.

Problem Statement:

In today's business environment, many businesses face challenges caused by disjointed organizational silos. In case of TechVision Enterprises, the absence of integration causes a number of key problems:

- Information Gap: Sales people lack visibility into stock balances resulting in unmet customer expectations.
- Revenue Loss: Calculations made manually in the billing department are prone to errors and mistakes, especially while calculating discounts and complex taxes such as GST.

- Lead Time Issues: The lag time from placing orders to picking goods from the warehouse causes delays that affect customer service.
- Misreporting: Lack of integration of Finance posting process causes old AR aging reports to be generated.

3. ORGANIZATIONAL STRUCTURE AND MASTER DATA

Effective SAP implementation is initiated through an effective organizational structure. TechVision Enterprises, as per the project requirements, was structured using the following enterprise structure parameters:

Enterprise Structure	Parameter Details
Company Code	TV01 (TechVision Enterprises Pvt. Ltd.)
Sales Organization	TV01 (Direct Sales India)
Distribution Channel	10 (Direct Sales)
Plant/Storage Location	TV01/0001 (Main Bengaluru Hub)
SAP Client	800

3.1 Master Data Configuration

Customer Master (XD01)

TechVision Buyer Corp (C-10001) has been set as 'Sold-to Party' and the following key parameters have been considered for the configuration:
 Reconciliation Account (140000): Automatically reconciles all trade receivable accounts
 Payment Terms (Z030): Net Credit Period is set for 30 Days

Material Master (MM01)

The material "Smart LED Monitor 27" (MAT-10045) has been created under the category of Finished Product (FERT). Following views have been configured in the material master Sales Org 1 (Delivering Plant) MRP 1 (Planning Parameters) Accounting: Valuation Class: 7920

Pricing Condition Records (VK11)

Following pricing condition records have been configured using RVAA01 as a pricing procedure to determine a reasonable pricing strategy:
 Base Price (PR00): 18,500.00
 Customer Discount (K007): 5%
 Output Tax (MWST/GST): 18%

4. THE ORDER TO CASH (O2C) BUSINESS CYCLE

The below are the transactions that will lead to the completion of the standard O2C process cycle in the SAP system. These are key points in the life cycle of customers.

Step 1: Pre Sales (VA11 & VA21)

The first step involves an Inquiry (VA11) received from the customer for 10 units of the Smart LED Monitor. It follows with Quotation (VA21) where the prices are locked along with the validity of dates (16.04.2026 to 30.04.2026) in total value amounting to INR 2,07,385.00.

Step 2: Sales Order Cycle (VA01)

On receipt of approval from the customer, we create a Standard Sales Order (OR) using the Quotation. An automatic credit check is carried out based on the customer's credit limit and also a checking for availability (ATP) in Plant TV01.

Step 3: Shipping (VL01N)

The creation of an Outbound Delivery (Type LF) initiates the shipment process. At this point, the WM module is triggered to create a Transfer order (LT0A) for picking from selected bins. Once the transfer order is completed (LT1A), Post Goods Issue (PGI) will be executed, which will lower the stock levels along with accounting entry of Dr. COGS / Cr. Inventory.

Step 4: Billing and Invoicing (VF01)

Upon actual shipping of the goods, the Billing Department prepares the Tax Invoice (Type F2). This is the formal document that will demand payment from the purchaser. When released to accounting (VF02), an account entry of: Dr. Accounts Receivable/ Cr. Sales Revenue+GST Payable is created.



5. TECHNICAL INTEGRATION AND STACKS USED

The strength of SAP software is in the integration of different functional areas.

In this case, the power of SAP can be seen in the "Seamless Flow" that takes place within a number of SAP modules:

SD-MM Integration: Stock depletion automatically done in Materials Management at PGI.

SD-FI Integration: Revenue and taxes posted immediately into GL after invoicing.

SD-WM Integration: Picking operations integrated with delivery document.

SAP Technology Components:

Components	Description
ERP System	SAP S/4HANA On-premise (2023)
Modules	SD, FI, MM, WM, PP
Report	VA05, VF05, FBL5N (Standard)
GST	Framework Indian

Sales Document Item: 10, Item category: AGN, Standard Item, Material: 200001583, Test mat

Condition...

Quantity: 1 EA, Net: 0,00 INR, Tax: 0,00

Condition Record, Analysis, Update

I.	CnTy	Name	Amount	CrCy	per	UoM	Condition Value	Curr.	Status	NumCCo	ATO/MTS Component	OUn	CConDe	Un
PROO		Price	1,00,000	INR		1 EA	0,00	INR			1	EA		1 EA

6. UNIQUE FEATURES

The following are some unique features of the project design that distinguish it from other normal academic projects:

Full tax calculation process: The project incorporates GST at 18%, which is consistent with India's current economic policies.

Document traceability: There is link between every document starting with an inquiry (10000001) all the way through to payment (1500000001).

Credit Control Checks: No sales orders can be raised for customers who are above the credit limit (FD32).

7. FUTURE SCOPE AND IMPROVEMENTS

Although the current O2C process works fine, it could be improved further by taking advantage of the advanced features offered by SAP:

- SAP C4C Integration: Migrating pre-sales to a cloud-based CRM system to improve lead management.
- RPA: Using RPA to automate the data entry of customer inquiries from their emails to SAP VA11.
- Advanced ATP (aATP): Using the advanced availability check in S/4HANA to promise delivery dates from several plants and logistics sites.
- EDI: Implementing electronic data interchange for B2B communication without the need for paperwork.

8. CONCLUSION

The SAP SD Capstone Project effectively highlights the automated O2C process of TechVision Enterprises. By combining the sales, logistics, and finance departments, the capstone project shows that an integrated ERP system reduces silos, guarantees tax compliance, and has the necessary scalability to accommodate the needs of a modern technology company. The final clearance of the AR account represents the successful closing of one complete business cycle.